How Well Do Globally Trained Sentinel-1 Flood Mapping Methods Transfer to Kenya? An Evaluation Against UNOSAT Rapid Mapping of the 2024 Long-Rains Floods

Craig Carlos Ouma*[1]

[1]Independent Researcher

October 10, 2026

Abstract

Kenya's March to May 2024 long rains brought flooding along the Tana River and across Nairobi and its neighbouring counties, and UNOSAT mapped it from optical satellite imagery, which cloud blocks during the rains. This paper measures how well Sentinel-1 radar flood mapping methods developed elsewhere transfer to these Kenyan landscapes, using UNOSAT's maps as an independent reference. Sentinel-1 scenes were processed to σ^0 in a way that reproduces the Sen1Floods11 training data to within 0.04 dB, and nine methods were compared: Otsu thresholds and log-ratio change detection, and U-Nets trained on Sen1Floods11, one of them with a dry-season image as extra input. No Kenyan data were used in training. The main U-Net matched published Sen1Floods11 results (all-water IoU 0.424 against 0.408). In riverine and rural settings, at Garissa and along 8,785 km^2 of the Tana and Galana rivers, it agreed with UNOSAT at an IoU of 0.347 (95 % block bootstrap interval 0.331 to 0.364), against 0.160 for the best classical method, and its flooded area was within about 1 % of UNOSAT's on two references. In the urban analysis of Nairobi, Kiambu and Machakos every method's interval overlapped, the U-Net reaching 0.221 and finding about a quarter of the mapped flood water. Built-up land was where every method failed, with flood-class F1 below 0.11 for all nine. Timing mattered as much as method: at Garissa the U-Net scored 0.215 against a reference three days after the radar pass and 0.581 against one nine days before it, because the river was still rising. The best extent model also placed only 3 to 14 % of UNOSAT's affected buildings inside its masks, while a fixed change-detection rule came within the spread expected from building data alone, so extent and exposure call for different products. Applied to every pass from April to June, the U-Net shows flood water on the middle Tana peaking at 535 km^2 on 20 May and near the delta about four weeks later.

1 Introduction

The March to May 2024 long rains were near or above average over most of Kenya, and at several Nairobi stations, at Thika and at Machakos they were the wettest season on record [8]. Flooding followed along the Tana River in Garissa and Tana River counties and across Nairobi and the neighbouring counties of Kiambu and Machakos, and landslides struck the towns of Mai Mahiu

*Correspondence: craigcarlos95@gmail.com. Code and data: github.com/craigouma

and Kijabe [29]. By 18 June the National Disaster Operations Centre counted 315 people killed, 188 injured and 38 missing, and more than 293,200 people displaced since 1 March [18].

The United Nations Satellite Centre (UNOSAT) activated rapid mapping for the event under code FL20240426KEN and published flood extents for Garissa and the village of Mororo opposite it, for the Tana and Galana rivers, and for the built-up area spanning Nairobi, Kiambu and Machakos [29]. Those maps were drawn from optical imagery, from Sentinel-2 and from very high resolution Pléiades scenes. Optical sensors cannot see through cloud, and cloud cover persists through the long rains. In the six days either side of UNOSAT's reference dates, the three Sentinel-2 acquisitions over Nairobi were 52 %, 89 % and 92 % cloud covered and the three over Garissa 80 %, 36 % and 100 %, as reported for the whole 110 km tile.

Sentinel-1 carries a C-band synthetic aperture radar (SAR) that images through cloud, by day and night, and its data are free. Since the loss of Sentinel-1B in December 2021, coverage has thinned: Garissa, for example, was imaged on a single track every 12 days through 2024. Open water is dark in SAR because it reflects the radar pulse away from the sensor, and this contrast is the basis of every SAR flood mapping method.

The methods available off the shelf were developed or trained elsewhere. A widely used labelled dataset for training deep learning models on Sentinel-1, Sen1Floods11 [3], spans eleven hand-labelled flood events in eleven countries, three of them African. Whether these methods hold up on Kenyan landscapes, where semi-arid floodplains, cropland, informal settlements and dense urban fabric each scatter radar differently, has not been measured against an independent reference. This paper measures it.

The contributions are four. First, classical and deep learning Sentinel-1 methods are run over the 2024 event and scored against UNOSAT's reference maps, separately for riverine and urban settings and broken down by land cover, with uncertainty from a spatial block bootstrap. Second, the sensitivity of that agreement to the number of days between the radar and optical acquisitions is measured directly, by scoring every Sentinel-1 pass within 24 days of each reference. Third, building exposure estimated from each method is compared with UNOSAT's counts of affected structures. Fourth, the method that performs best in riverine settings is applied to every pass over the lower Tana between April and June 2024, as an application with its measured error attached.

The paper is the companion to an earlier study in this series on satellite indicators of drought and food insecurity in northern Kenya [20], and covers the opposite hydrological hazard with a second sensor type.

2 Background

2.1 SAR flood mapping

The simplest method classifies a single post-event image by a backscatter threshold. Because the threshold depends on sensor, incidence angle, wind and land cover, it is usually fitted per scene, most often with Otsu's method [19], which chooses the threshold that maximises the separation of a bimodal histogram. A histogram over a whole scene is rarely bimodal when water covers a small fraction of it, so split-based approaches fit the threshold only on tiles that contain both classes [11, 4]; the same idea underlies fully automated processing chains for Sentinel-1 [28, 12].

Change detection compares the flood image with a reference image from dry conditions, usually as a log ratio, and labels as flooded the pixels that became darker [2, 6]. It removes permanently dark surfaces such as tarmac, sand and smooth bare soil, which single-image thresholds confuse with water, at the cost of needing a suitable reference image from the same orbit [6]. Terrain descriptors

such as Height Above the Nearest Drainage (HAND) [23, 17] are used to remove water-like pixels on slopes and hilltops where flooding is implausible.

The Copernicus Emergency Management Service now runs a global flood monitoring product on every Sentinel-1 acquisition, as an ensemble of three independently developed algorithms [25, 10].

2.2 Sen1Floods11

Sen1Floods11 [3] provides 4,831 chips of 512 by 512 pixels at 10 m from eleven flood events, each with Sentinel-1 VV and VH backscatter processed in Google Earth Engine. 446 chips carry hand-checked labels and are split into training, validation and test sets, with the Bolivia event held out as an unseen test; the remaining 4,385 carry weak labels from automated classifications of Sentinel-1 or of coincident Sentinel-2. In the original paper the best flood-water results came from a fully convolutional network trained on the Sentinel-2-derived weak labels, with a mean per-chip IoU of 0.339 for flood water and 0.408 for all water on the test set, against 0.285 and 0.359 for Otsu thresholding of VH. Later work has reported Sentinel-1-only results on the same benchmark with other architectures and loss functions [1, 9]; the highest published Sen1Floods11 numbers come from optical foundation models [7, 27], which do not use the radar channels.

2.3 The urban SAR problem

In built-up areas, walls and the ground form corner reflectors, and floodwater between buildings strengthens this double-bounce return instead of darkening it [13, 14]. Radar shadow and layover behind tall structures add dark and bright artefacts unrelated to water. Floodwater in towns is therefore often invisible to backscatter thresholds, and approaches that work there rely on interferometric coherence or on combining SAR with an elevation model of the urban surface [15]. The Sen1Floods11 validation data include no urban flood events [3].

3 Data

3.1 Reference maps and their limits

UNOSAT published twelve products for activation FL20240426KEN on the Humanitarian Data Exchange under CC BY-SA [29]. Every product page links the same cumulative shapefile package, and the products resolve into fewer independent analyses than pages (Table 1). The four urban flood products are views of one Pléiades analysis of 1 May 2024 with a single 580 km^2 extent, flood layer and structure layer; their reported structure counts (3,387, 3,663, 4,246 and 1,815) sum to exactly the 13,111 points in that layer. They are treated here as one area of interest (AOI). The Sentinel-2 analysis of Garissa on 29 April has no flood layer of its own in the package, and its flood water is taken from the Tana and Galana product derived from the same image, clipped to the Garissa extent; this gives 27.9 km^2 against the 26 km^2 UNOSAT reports. The Tana and Galana product covers 70,874 km^2 and has no product page of its own. The national NOAA-VIIRS composite (375 m pixels), the Mai Mahiu landslide product and three cropland subsets of the urban analysis are not used. Figure 1 locates the areas used.

These maps are an independent reference, not ground truth. Each product states that it is a preliminary analysis not yet validated in the field. They were digitised from optical images by analysts working to a deadline, at resolutions from 0.5 m (Pléiades) to 10 m (Sentinel-2), and they record water as seen from above: flooding under tree canopy, or water so shallow or turbid that it is spectrally indistinguishable from wet soil, may be missed or mapped inconsistently. They were

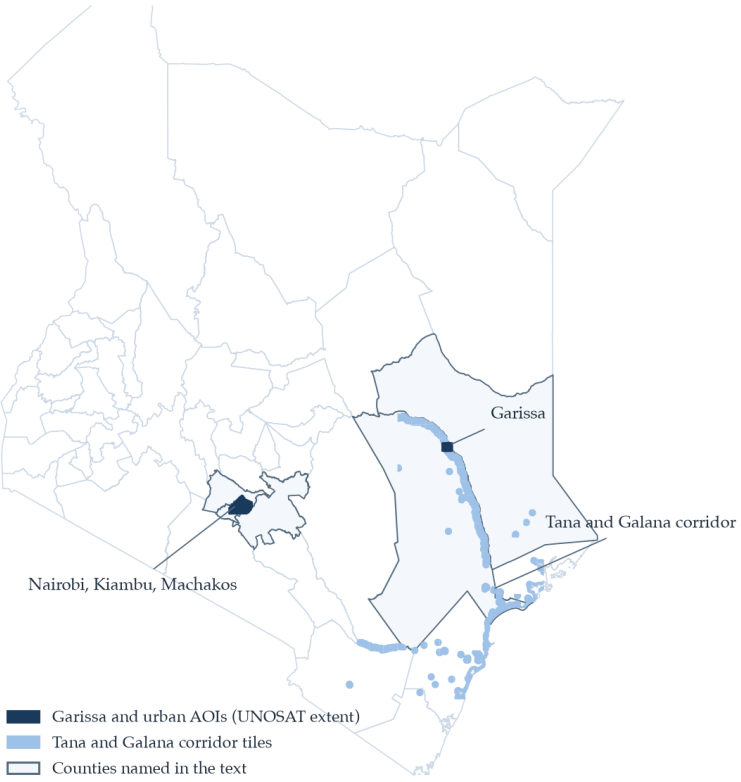


Figure 1: Areas evaluated: the UNOSAT analysis extents at Garissa and in the Nairobi, Kiambu and Machakos area, and the 70 tiles of the Tana and Galana corridor, over Kenya's county boundaries.

also acquired on different days from the radar. Disagreement between a SAR method and these maps therefore combines errors of the method, errors of the reference and real change between the two acquisitions. Every UNOSAT flood layer used here is classed as new flood water, separate from the river, and UNOSAT's cloud-obstruction polygons, where its analysts could not see the ground, are removed from the evaluation domain.

3.2 Sentinel-1 imagery

Sentinel-1 Interferometric Wide swath Ground Range Detected (GRD) scenes in VV and VH polarisation were read from Microsoft Planetary Computer [22]. Its radiometrically terrain corrected collection now requires an account, so the GRD scenes were processed here: thermal noise removal, calibration to σ^0 from the product's annotation tables, and geocoding from the annotation geolocation grid onto a 10 m UTM zone 37S grid, with each pixel shifted in range by $(h - h_{\mathrm{ann}})/\tan\theta$ for its height h above the ellipsoid, where h_{ann} is the terrain height the GRD was projected to and θ the incidence angle. Heights come from the Copernicus DEM GLO-30 [5] converted from EGM2008 to ellipsoidal height (the geoid lies 27 m below the ellipsoid at Garissa and 16 m at Nairobi). No radiometric terrain flattening is applied. This reproduces the processing behind the Sen1Floods11 chips. On 18 hand-labelled chips from nine events, processing the original scenes this way reproduced the Sen1Floods11 values with a median bias between -0.02 and $+0.04$ dB and a median correlation of 0.97, with zero or one pixel of offset for all events except one, so models trained on

Table 1: Reference analyses used. Day gap is the Sentinel-1 date minus the reference image date for the pass used in the headline results.

AOI	Group	Reference image	Sentinel-1 pass	Day gap	Flood water (km^2)
Garissa, 29 April	Riverine	Sentinel-2, 29 Apr	26 Apr	−3	27.9
Garissa, 5 May	Riverine	Pléiades, 5 May	8 May	+3	21.7
Tana and Galana, 70 tiles	Riverine	Sentinel-2, 29 Apr	26 Apr, 1 May	−3, +2	361.0
Nairobi, Kiambu, Machakos	Urban	Pléiades, 1 May	1 May	0	59.7

Sen1Floods11 see Kenyan inputs from the same radiometric domain.

For each AOI the post-event pass is the one closest in time to the reference image, and the pre-event pass is the latest January or February 2024 acquisition from the same relative orbit. A pass is all slices of one datatake that image the AOI, mosaicked. Garissa is covered by a single track (relative orbit 159, ascending) and the Nairobi area by two (relative orbits 57 and 130, both ascending); the headline Nairobi pass is on orbit 57, on the day of the reference.

3.3 Training data

Sen1Floods11 v1.1 [3] was downloaded from its public bucket. It holds eleven hand-labelled events and a twelfth, in Colombia, with weak labels only. Three of the eleven are African: Ghana (53 hand-labelled chips), Nigeria (18) and Somalia (26). Somalia, the Juba and Shabelle floods of May 2018, is the only East African event, so the transfer to Kenya is out of region for most of the training data but not all of it. No Kenyan data, and no UNOSAT data, were used in training.

3.4 Ancillary layers

Permanent water is taken from JRC Global Surface Water occurrence [21], land cover from ESA WorldCover 2021 at 10 m [30], HAND from the global GLO-30 HAND layer derived from the Copernicus DEM, buildings from Google Open Buildings v3 [26] and county boundaries from the Kenya common operational dataset on the Humanitarian Data Exchange, as in the companion study [20].

4 Methods

4.1 Classical methods

Four classical methods were run, each on VV and on VH. **Otsu thresholding** labels as water every pixel of the 3×3 focal mean of post-event backscatter below a threshold fitted with Otsu's method [19]. Following the Sen1Floods11 reference classification [3], the threshold is fitted on the pooled histogram of the most heterogeneous 1 km tiles (the top 10 %, at least five), since a histogram dominated by dry land is not bimodal. **Change detection** uses the difference in dB between the smoothed post-event and pre-event images, which is the log ratio of the two, and labels as flooded the pixels that darkened by more than a threshold. Two thresholds are compared: Otsu's, fitted the same way and capped at 0 dB so that brightening is never labelled as flooding, and a fixed −3 dB.

4.2 U-Nets trained on Sen1Floods11

A four-level U-Net [24] with 32 base channels and group normalisation was trained on Sentinel-1 VV and VH chips. Training follows the original paper's setting that gave its best flood-water results: the hand-labelled training split plus the weakly labelled chips with labels derived from Sentinel-2, random 256×256 crops with flips, cross entropy weighted 1:8 towards water, AdamW with learning rate 5×10^{-4} and weight decay 10^{-2}, and cosine annealing with warm restarts. Inputs are scaled exactly as in the reference code. The Bolivia event is held out entirely, including the weakly labelled Bolivia chips that the public bucket contains, and the checkpoint with the best mean per-chip IoU on the validation split is kept. Training ran for 150 epochs on a single Kaggle T4 GPU.

Sen1Floods11 has no pre-event imagery. To test whether a model that sees the dry-season image does better, pre-event chips were built for the hand-labelled set with the processing described above, from the latest pass on the same relative orbit at least 30 days before each event; 419 of the 446 chips have one. A U-Net with four input channels (post and pre VV and VH) was trained on the hand-labelled chips that have a pair, alongside a control with the post-event channels only, trained on the same chips. The two differ only in their inputs.

4.3 Post-processing

Every method's output is post-processed identically. Pixels with JRC water occurrence above 80 % are removed from the prediction and from the evaluation domain. A second variant also sets to dry every pixel more than 20 m above the nearest drainage. The JRC rule removes almost nothing along the Tana at Garissa, where occurrence peaks at 72 % because the channel migrates and carries sediment.

4.4 Evaluation

Agreement is measured for the flood class as intersection over union (IoU), F1, precision and recall over the evaluation domain: the UNOSAT analysis extent minus cloud obstruction, permanent water, and pixels without radar data. AOIs are grouped as riverine and rural (Garissa and the Tana and Galana corridor) and urban and peri-urban (the Nairobi, Kiambu and Machakos analysis), and the groups are reported separately throughout because SAR behaves differently in built-up areas. Within a group, confusion counts are pooled across AOIs. The 95 % intervals come from 1,000 bootstrap resamples of 1 km blocks rather than pixels, since neighbouring pixels are strongly correlated. Results are also broken down by ESA WorldCover class.

The Tana and Galana product cannot be evaluated over its whole 70,874 km^2 without scoring mostly dry land far from any river. It is evaluated along a corridor defined independently of the reference: land within 5 km of any 300 m cell where JRC water occurrence reaches 50 %, cut into 20 km tiles and excluding the Garissa extent, which is scored on its own. The corridor covers 8,785 km^2 and contains 361 of the 469 km^2 of flood water in that product.

To measure the effect of the time between the radar and optical images, every Sentinel-1 pass within 24 days of each point AOI's reference date was run through the same methods and scored against the same reference.

Building exposure is the number of Open Buildings v3 centroids with confidence of at least 0.70 inside each flood mask and the evaluation domain. UNOSAT counted affected structures on its own imagery, not on Open Buildings, so the number of Open Buildings centroids inside UNOSAT's own flood extent is reported as well, to separate disagreement in the building data from disagreement in the flood extent.

5 Results: Sen1Floods11 test performance

Table 2 scores every method on the Sen1Floods11 test split and on the held-out Bolivia event, with the original paper's convention (the mean of per-chip IoUs) and with IoU pooled over all pixels. The U-Net trained on hand-labelled and Sentinel-2-derived weak labels reaches a mean per-chip IoU of 0.424 for all water and 0.358 for flood water on the test split, and 0.446 and 0.392 on Bolivia. The original paper's best model, trained on the same labels, reported 0.408 and 0.339 on the test split and 0.316 and 0.274 on Bolivia [3]. The model is therefore in line with, and slightly above, the published range, and its Kenyan results can be read as those of a competent Sen1Floods11 model. Its best validation score came at epoch 121 of 150.

Otsu thresholding of VH, fitted on heterogeneous tiles of each chip, scores 0.224 for flood water on the test split and 0.387 on Bolivia, against 0.285 and 0.324 reported for the original per-event threshold. Change detection, which can only be scored on the 86 test and 9 Bolivia chips that have a pre-event image, scores between 0.10 and 0.12 on the test split. The pre-event images were drawn from one to five months before events that mostly occurred within long wet seasons, and many are probably not dry; the same method has a dry-season reference in Kenya.

On the paired chips, giving the U-Net the pre-event image changes flood-water IoU from 0.332 to 0.337 on the test split, within noise, and from 0.322 to 0.410 on Bolivia, the one event no model saw in training.

Table 2: Sen1Floods11 results, flood water. Mean is the mean of per-chip IoUs (the original paper's convention); pooled is IoU over all pixels. Paired rows use only chips with a pre-event image (86 test, 9 Bolivia).

	Test		Bolivia	
Method	Mean	Pooled	Mean	Pooled
Bonafilia et al. (2020), Otsu VH	0.285		0.324	
Bonafilia et al. (2020), S2 weak labels	0.339		0.274	
Otsu, VH	0.224	0.194	0.387	0.503
U-Net, post, hand and weak labels	0.358	0.570	0.392	0.680
Paired chips				
Change, −3 dB, VV	0.118	0.235	0.222	0.398
U-Net, post, hand labels only	0.332	0.516	0.322	0.500
U-Net, pre and post, hand labels only	0.337	0.514	0.410	0.618

6 Results: transfer to Kenya

6.1 Riverine and urban agreement

Table 3 gives agreement with UNOSAT pooled within each group, and Figure 2 plots it. In riverine and rural settings the U-Net trained on hand-labelled and weak labels agrees best with the reference, with an IoU of 0.347 (95 % interval 0.331 to 0.364), roughly double the best classical method, change detection on VV at a fixed −3 dB (0.160, 0.150 to 0.172). The intervals do not overlap. Single-image Otsu thresholds and Otsu-fitted change detection do worst, with precision between 0.07 and 0.15: they label as water far more than UNOSAT mapped.

In the urban and peri-urban analysis the ordering compresses. The three U-Nets score between 0.217 and 0.228 and the best classical method, Otsu on VH, 0.192 (0.161 to 0.225); the intervals

overlap, and no method can be said to be better than the others. The U-Nets' precision there is 0.62 to 0.65 and their recall about 0.26: what they label as water in Nairobi is usually water in the reference, but they find about a quarter of the flood water UNOSAT mapped. Methods that reach higher recall do so by labelling large parts of the city as water; Otsu on VV finds 82 % of the reference at a precision of 0.16. The automatic thresholds also break down in the city: Otsu's threshold on VV falls at −7.6 dB, within the range of ordinary dry land: the brightest mode of the urban histogram is buildings, and the fitted threshold falls between buildings and everything else. The change-detection fits land at −0.65 dB for VV and +0.22 dB for VH, where the 0 dB cap applies, so almost any darkening counts as flooding.

Table 3: Agreement with UNOSAT flood water, pooled within each group, with 95 % intervals from a 1 km block bootstrap. P is precision and R recall. Riverine and rural: Garissa (two references) and 70 Tana and Galana corridor tiles. Urban and peri-urban: the Nairobi, Kiambu and Machakos analysis.

	Riverine and rural			Urban and peri-urban		
Method	IoU (95 % CI)	P	R	IoU (95 % CI)	P	R
Otsu, VV	0.082 (0.077–0.088)	0.10	0.35	0.158 (0.137–0.180)	0.16	0.82
Otsu, VH	0.116 (0.109–0.125)	0.15	0.34	0.192 (0.161–0.225)	0.41	0.27
Change, Otsu, VV	0.064 (0.061–0.069)	0.07	0.41	0.175 (0.145–0.206)	0.31	0.29
Change, Otsu, VH	0.064 (0.060–0.069)	0.07	0.43	0.164 (0.140–0.189)	0.23	0.37
Change, −3 dB, VV	0.160 (0.150–0.172)	0.27	0.28	0.151 (0.115–0.189)	0.62	0.17
Change, −3 dB, VH	0.155 (0.144–0.168)	0.21	0.37	0.165 (0.130–0.202)	0.57	0.19
U-Net, post	0.347 (0.331–0.364)	0.53	0.50	0.221 (0.171–0.273)	0.62	0.26
U-Net, post, hand labels only	0.212 (0.200–0.225)	0.29	0.45	0.217 (0.167–0.269)	0.62	0.25
U-Net, pre and post, hand labels only	0.277 (0.262–0.293)	0.39	0.49	0.228 (0.175–0.282)	0.65	0.26

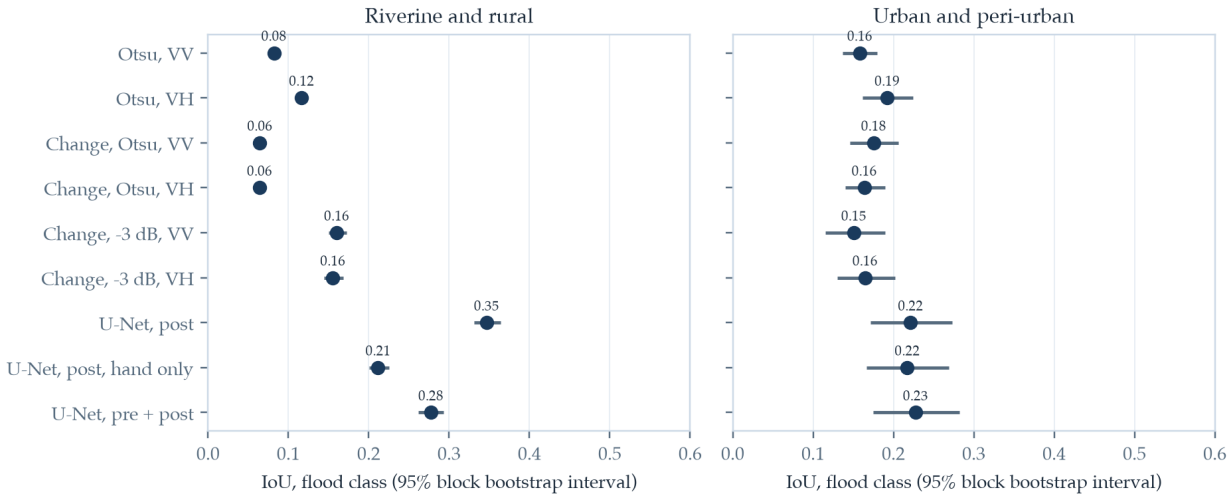


Figure 2: IoU for the flood class by method and group, with 95 % intervals from resampling 1 km blocks.

6.2 Individual areas

The pooled riverine figure is dominated by the Tana and Galana corridor, which holds 360 of the 410 km^2 of riverine reference flood water. Within it the U-Net's IoU is 0.344 and change detection's 0.147. Change detection's errors there are mostly commission in shrubland, grassland and tree cover away from the channels: patches that darkened between February and late April but that UNOSAT did not map as flood water. They may be rain-filled pans and depressions on the semi-arid plain, flooding under vegetation that an optical image cannot see, or false alarms, and the data here cannot separate the three. The U-Net's flooded area over the corridor, 356 km^2, is within about 1 % of the reference's 360 km^2.

At Garissa the two references give very different pictures. Against the Pléiades map of 5 May, matched by a radar pass three days later, the U-Net reaches an IoU of 0.544 with precision and recall both near 0.70, and its flooded area, 21.5 km^2, is within about 1 % of the reference's 21.7 km^2. Change detection on VV reaches 0.487. Against the Sentinel-2 map of 29 April, matched by a pass three days earlier, every method falls to between 0.10 and 0.22, with recall below 0.25. Figure 3 shows the 5 May case.

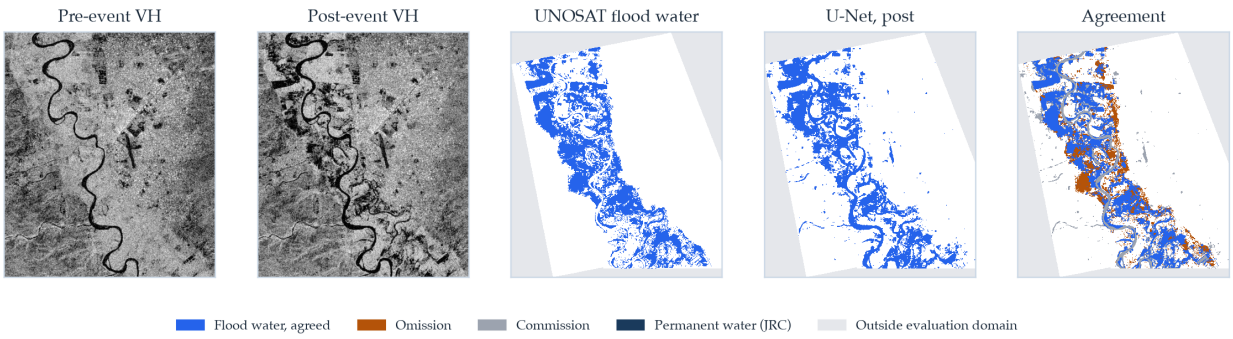


Figure 3: Garissa and Mororo, reference of 5 May 2024. From left: pre-event VH (26 February), post-event VH (8 May), UNOSAT flood water, the U-Net's flood water, and agreement (blue), omission (amber) and commission (grey).

6.3 Sensitivity to the day gap

The Garissa contrast is a timing effect, which scoring every Sentinel-1 pass within 24 days of each reference shows directly (Figure 4). Against the 29 April reference, change detection on VV scores 0.163 with the 26 April pass and 0.443 with the 8 May pass, nine days after the reference. Against the 5 May reference it scores 0.487 with the pass three days later and 0.174 with the pass nine days earlier. The radar mapped 7.3 km^2 of flood water on 26 April and 16.7 to 18.1 km^2 on 8 May: the Tana was still rising at Garissa in late April, and the 29 April optical image caught a larger flood than the radar three days before it. Passes 15 or more days from either reference score below 0.07 with change detection. The U-Net follows the same pattern at a higher level: 0.215 with the 26 April pass and 0.581 with the 8 May pass against the 29 April reference, and 0.544 with the pass three days after the 5 May reference. In Nairobi, which two tracks cover, agreement exists only on the day of the reference: 0.221 for the U-Net and 0.151 for change detection on 1 May, against 0.006 to 0.035 for every pass between 5 and 24 days away on either side. Urban flood water had drained within five days. Otsu on VH keeps an IoU near 0.13 on some distant passes because it labels water that is unrelated to the flood.

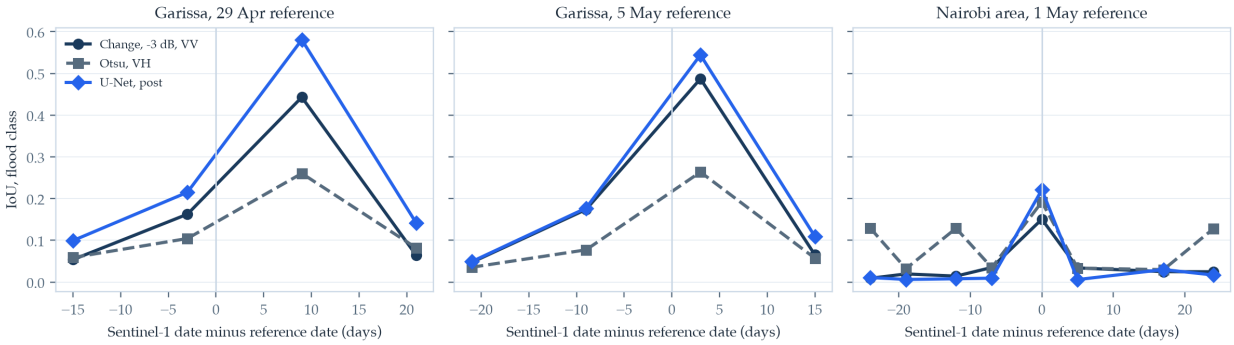


Figure 4: IoU against the number of days between the Sentinel-1 pass and the reference image, for every pass within 24 days of each reference.

6.4 Land cover

Figure 5 breaks agreement down by ESA WorldCover class. Built-up land is the class where every method fails, in both groups: F1 for the flood class is below 0.11 for all nine methods. UNOSAT itself mapped little flood water on land WorldCover classes as built-up (0.7 km^2 in the riverine AOIs and 2.7 km^2 in the urban one, against 30.5 km^2 on grassland in the urban analysis, mostly along river corridors), so these scores rest on small areas, but the failure is uniform. The best class is cropland for the U-Nets (F1 0.68 riverine, 0.54 urban) and herbaceous wetland for the classical methods in riverine settings (0.53 to 0.60). For the U-Net, riverine shrubland is the weakest vegetated class (F1 0.21).

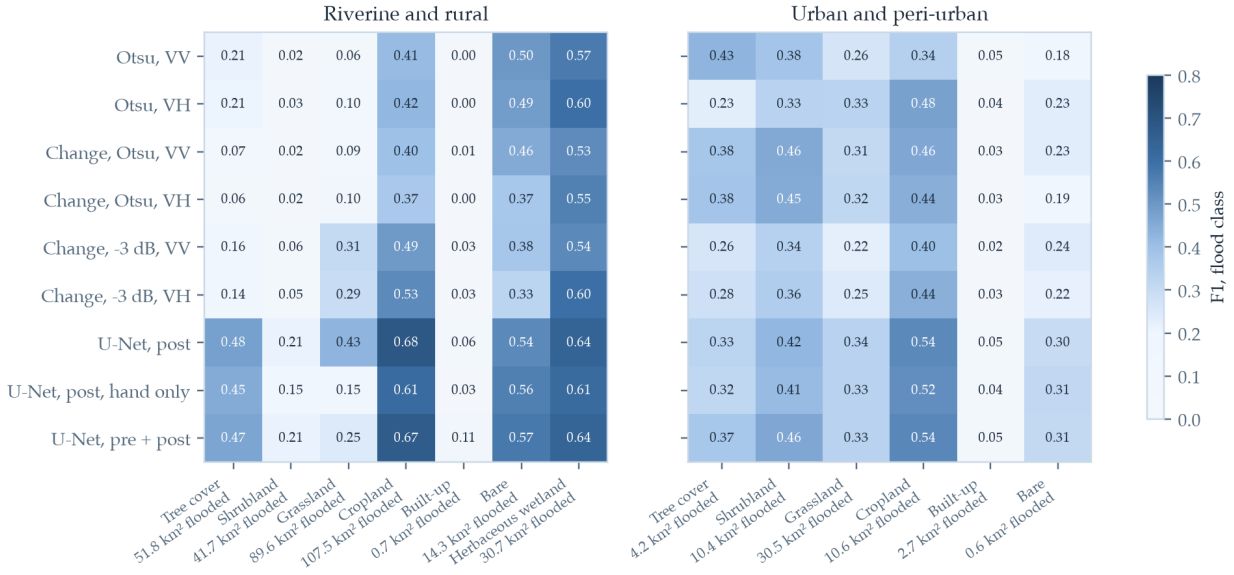


Figure 5: F1 for the flood class by ESA WorldCover class and method. Labels under each class give the reference flood water it contains.

6.5 HAND mask

Setting to dry every pixel more than 20 m above the nearest drainage changes IoU by at most 0.008 for any method in either group. Along the lower Tana and in Nairobi's river valleys nearly all flood water, and nearly all false alarms, lie within 20 m of drainage, so the mask has little to remove.

7 Exposure estimates against UNOSAT counts

UNOSAT counted 3,440 potentially affected buildings at Garissa and Mororo on 29 April, 3,377 on 5 May, and 13,111 affected structures across the urban analysis on 1 May, of which 12,985 fall inside the evaluation domain here [29]. Those counts were digitised by analysts on very high resolution imagery. Table 4 sets against them the number of Google Open Buildings v3 centroids [26] inside each method's flood mask, and inside UNOSAT's own flood extent. The first row separates the two sources of disagreement: even with UNOSAT's flood extent, Open Buildings gives 0.72, 1.26 and 0.67 times UNOSAT's counts, so differences of about 30 % in either direction come from the building data alone, and agreement closer than that should not be expected.

Table 4: Buildings in flood masks: Open Buildings v3 centroids (confidence at least 0.70) inside each mask and the evaluation domain, with the ratio to UNOSAT's structure count in brackets. UNOSAT counts in the domain: 3,440 (Garissa, 29 April), 3,377 (Garissa, 5 May) and 12,985 (urban).

Mask	Garissa, 29 Apr	Garissa, 5 May	Nairobi area, 1 May
Open Buildings in UNOSAT flood extent	2,481 (0.72)	4,241 (1.26)	8,736 (0.67)
U-Net, post	116 (0.03)	488 (0.14)	757 (0.06)
U-Net, pre and post	145 (0.04)	852 (0.25)	886 (0.07)
Change, −3 dB, VV	1,379 (0.40)	2,557 (0.76)	7,246 (0.56)
Change, −3 dB, VH	1,919 (0.56)	3,417 (1.01)	10,166 (0.78)
Otsu, VH	797 (0.23)	327 (0.10)	1,484 (0.11)

The method that maps flood extent best is among the worst at finding flooded buildings. The U-Net trained on hand-labelled and weak labels puts 116, 488 and 757 buildings in its masks, between 0.03 and 0.14 times UNOSAT's counts. Its masks follow open floodplain and almost never enter settlements, which is consistent with its F1 of 0.05 to 0.06 on built-up land and with a training set that contains no urban floods. Change detection with a fixed −3 dB threshold on VH comes closest to UNOSAT's counts, at 0.56, 1.01 and 0.78 times. It does so with low pixel-level agreement in Nairobi (IoU 0.165), so the closeness of its count is partly a matter of commission and omission cancelling; it nevertheless shows that backscatter decreases in and around Kenyan settlements carry information about flooded buildings that the U-Net discards. In the urban analysis, Otsu on VV and Otsu-fitted change detection, whose thresholds fail there (Section 6), overcount by factors of 4.7 to 9.6, while Otsu on VH undercounts at 0.11 (Figure 6).

8 Lower Tana flood time series

The U-Net trained on hand-labelled and weak labels, the best method in riverine settings, was applied to every Sentinel-1 pass between 1 April and 30 June 2024 over the 38 corridor tiles in Garissa and Tana River counties and the Garissa extent, 273 tile passes in all. Two tracks cover this region. Relative orbit 159 images 5,755 km^2 of the corridor in Tana River County and 1,682 km^2

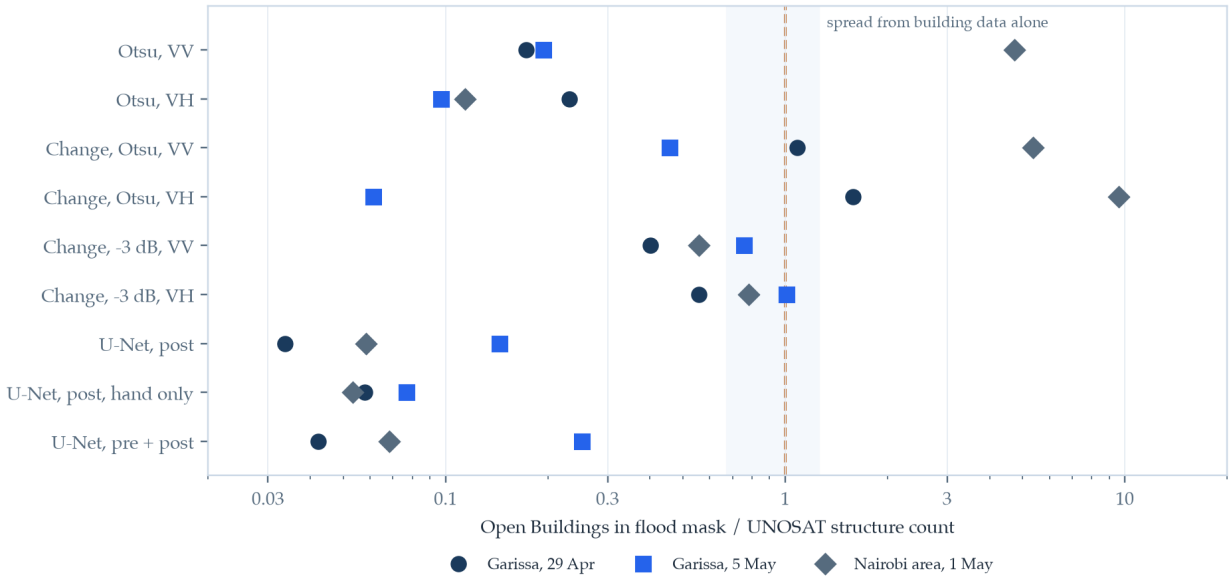


Figure 6: Open Buildings centroids inside each method's flood mask divided by UNOSAT's structure count, on a logarithmic scale. The dashed line is equality; the shaded band spans the ratios obtained with UNOSAT's own flood extent, the disagreement that comes from the building data alone.

in Garissa County every 12 days; relative orbit 57 images 205 km^2 of the lower river near the delta. The 1 June acquisition on orbit 159 and the 13 May acquisition on orbit 57 are not in the archive. Flood water is reported per county and per orbit (Figure 7), and areas are comparable only within an orbit, since each images a different part of the river. There is no reference for most of this area, so these are estimates with the riverine accuracy of Section 6 attached: an IoU of 0.352 against the 29 April reference on the 37 corridor tiles the series covers, with flooded area within about 1 % of the reference on the 29 April Tana product and on the 5 May Garissa map.

On the middle Tana (orbit 159), water in Tana River County rose from 236 km^2 on 2 April to 330 km^2 on 26 April and 482 km^2 on 8 May, peaked at 535 km^2 on 20 May, and fell to 343 km^2 by 25 June. The 2 April figure includes the river channel itself, which JRC records as water less than 80 % of the time and which the permanent-water rule therefore keeps, together with any early-season flooding; the rise from that baseline is the signal. Near the delta (orbit 57) water kept rising after the middle river had peaked, from 19 km^2 on 7 April and 63 km^2 on 1 May to 122 km^2 on 18 June, before falling to 96 km^2 on 30 June. The lower river peaked about four weeks after the middle river, as expected for a flood wave moving downstream through a long, flat floodplain. On 18 June, OCHA reported 46 displacement camps in Tana River County hosting 34,205 people, the most of any county, against three sites hosting about 4,140 people in Garissa County [18].

The Garissa County side of the corridor carried far less water, 8 to 14 km^2 throughout, because the floodplain at Garissa lies mostly on the Tana River County bank, where Mororo is. The U-Net also underestimated flooding at Garissa on 26 April, when the river was still rising (Section 6), and its masks avoid built-up land (Section 7), so flooding in Garissa town is likely understated in this series.

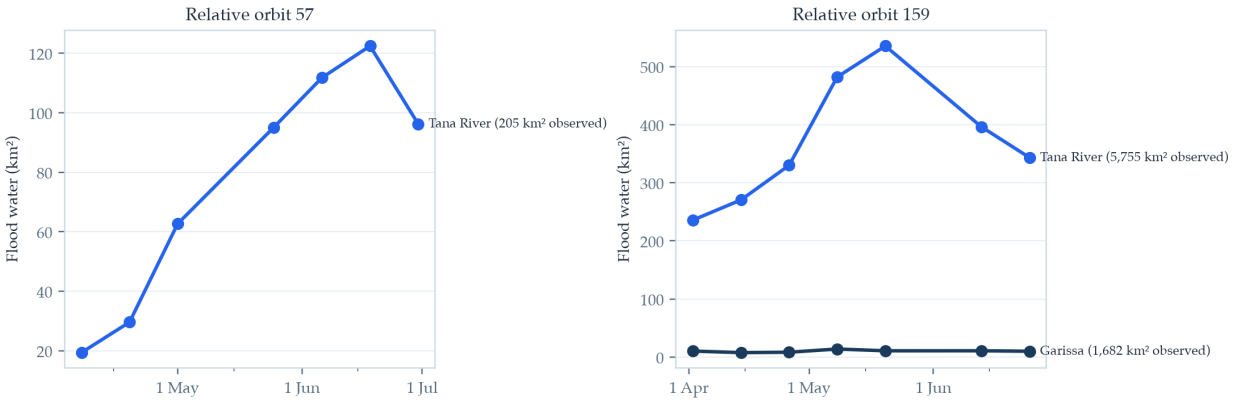


Figure 7: Flood water mapped by the U-Net along the lower Tana corridor in Garissa and Tana River counties for every Sentinel-1 pass from April to June 2024, by relative orbit. Labels give the area each orbit observes in each county.

9 Implications for flood monitoring in Kenya

The first implication is that a model trained on Sen1Floods11 transfers to Kenyan floodplains well enough to use, and better than the classical methods that are easier to deploy. Along the lower Tana it agreed with an independent optical reference about twice as well as the best threshold or change rule, its total flooded area was within about 1 % of UNOSAT's on two references, and its agreement on the one reference matched in time and resolution (IoU 0.544 at Garissa on 5 May) is comparable to its score on Sen1Floods11's own test chips. The training set contains one East African event among eleven, so this is mostly out-of-region transfer, and it held. A Kenyan monitoring system that receives Sentinel-1 scenes every 12 days could produce cloud-independent flood extent along the Tana and its peers with a public model and public data, which is what Section 8 does for April to June 2024.

The second is that the same model should not be used to count affected people or buildings. Its masks avoid settlements, so building exposure computed from them would understate impact by an order of magnitude in both Garissa and Nairobi. Change detection with a fixed threshold, despite its lower pixel agreement, came within the range expected from the building data alone. A practical system would use one product for extent and another, or explicit treatment of built-up areas, for exposure; a single headline accuracy figure would hide this.

The third is that radar alone does not resolve urban flooding in Nairobi. The methods that keep false alarms in check, the U-Net among them, find about a quarter of the flood water UNOSAT mapped from Pléiades on the same day, and none is clearly better than the others. Urban flooding in Nairobi and its neighbours is short-lived and concentrated along narrow river corridors and among buildings, where backscatter brightens instead of darkening. Interferometric coherence and urban elevation models [15] are the established ways forward, and the Sen1Floods11 training data contain no urban events from which a model could learn the urban signature.

The fourth concerns the reference itself. At Garissa, the three days between the radar and the optical image while the river was rising changed agreement more than the choice of method did. Rapid mapping products are made from whichever image is available first, and an evaluation that ignores the acquisition dates would have attributed a timing effect to the method. Any routine comparison between SAR products and rapid mapping in Kenya should report the day gap, and during a rising flood a pass a few days after the optical image can be a better match than one a

few days before it.

10 Limitations

The reference maps are preliminary rapid analyses, not field-validated ground truth, and they were made from optical imagery by analysts working quickly. They can miss water under canopy and shallow water that looks like wet soil, and the Tana and Galana analysis in particular was mapped at 10 m over 70,874 km^2. Part of every method's disagreement with them is the reference's error, which this design cannot separate from the method's.

The radar and optical images were acquired on different days, by three days at Garissa, two to three days along the Tana and on the same day only in Nairobi. Section 6 shows that three days during a rising flood are enough to change IoU by a factor of almost three.

There are few independent analyses. The riverine group rests on two reference dates at Garissa and one Sentinel-2 product along the Tana and Galana, and the urban group on a single Pléiades analysis. The 70 corridor tiles are not 70 independent tests: they share one reference image and, for 57 of them, one radar pass. The block bootstrap accounts for spatial correlation within these analyses but not for the small number of analyses.

The corridor was defined from JRC water rather than from the reference, but it still restricts the Tana evaluation to land within 5 km of persistent water. Flooding farther from the rivers, and false alarms there, are not scored.

SAR physics limits what any backscatter method can see in towns, and none of the methods here uses coherence. The urban result describes backscatter methods; coherence-based methods were not tested.

The U-Net was trained with the original paper's labels and schedule but a different architecture, and on one seed; its Kenyan scores would vary somewhat between training runs. The pre-event U-Net and its control were trained on about 240 chips, eighteen times fewer than the main model, so the comparison between them is informative but neither is a competitive model on its own.

Building counts use Open Buildings v3 centroids above a single confidence threshold, while UNOSAT counted structures on its own imagery. The comparison is approximate by construction, and the row computed with UNOSAT's own flood extent shows how approximate.

No result here was validated in the field.

Reproducibility statement

Every input is public and was obtained without an account: UNOSAT products from the Humanitarian Data Exchange, Sentinel-1 GRD scenes, the Copernicus DEM, JRC Global Surface Water and ESA WorldCover from Microsoft Planetary Computer, Sen1Floods11 from its public bucket, GLO-30 HAND from ASF on Amazon Web Services, and Google Open Buildings v3 from its public bucket. The code, the notebook that regenerates every number and figure, the trained U-Net weights, the Kaggle kernels that trained them, the rasterised references, every method's flood mask for every AOI, and the lower Tana flood rasters are at `https://github.com/craigouma/kenya-floods-sar-2024` under the MIT licence for code and CC BY 4.0 for derived data, with UNOSAT's data and their derivatives under CC BY-SA. Every choice the analysis plan left open is recorded, with its reason, in the repository's `DECISIONS.md`. This is independent research; it was not produced, reviewed, commissioned or endorsed by UNOSAT, UNITAR, ESA, the Copernicus programme or any Kenyan agency.

References

[1] Bai, Y., Wu, W., Yang, Z., Yu, J., Zhao, B., Liu, X., Yang, H., Mas, E., and Koshimura, S. (2021). Enhancement of detecting permanent water and temporary water in flood disasters by fusing Sentinel-1 and Sentinel-2 imagery using deep learning algorithms: demonstration of Sen1Floods11 benchmark datasets. *Remote Sensing*, 13(11), 2220.

[2] Bazi, Y., Bruzzone, L., and Melgani, F. (2005). An unsupervised approach based on the generalized Gaussian model to automatic change detection in multitemporal SAR images. *IEEE Transactions on Geoscience and Remote Sensing*, 43(4), 874-887.

[3] Bonafilia, D., Tellman, B., Anderson, T., and Issenberg, E. (2020). Sen1Floods11: a georeferenced dataset to train and test deep learning flood algorithms for Sentinel-1. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 835-845.

[4] Chini, M., Hostache, R., Giustarini, L., and Matgen, P. (2017). A hierarchical split-based approach for parametric thresholding of SAR images: flood inundation as a test case. *IEEE Transactions on Geoscience and Remote Sensing*, 55(12), 6975-6988.

[5] European Space Agency and Airbus (2022). *Copernicus DEM*. `https://doi.org/10.5270/ESA-c5d3d65`

[6] Hostache, R., Matgen, P., and Wagner, W. (2012). Change detection approaches for flood extent mapping: how to select the most adequate reference image from online archives? *International Journal of Applied Earth Observation and Geoinformation*, 19, 205-213.

[7] Jakubik, J., Roy, S., Phillips, C. E., Fraccaro, P., Godwin, D., Zadrozny, B., Szwarcman, D., Gomes, C., Nyirjesy, G., Edwards, B., Kimura, D., Simumba, N., Chu, L., Mukkavilli, S. K., Lambhate, D., Das, K., Bangalore, R., Oliveira, D., Muszynski, M., Ankur, K., Ramasubramanian, M., Gurung, I., Khallaghi, S., Li, H., Cecil, M., Ahmadi, M., Kordi, F., Alemohammad, H., Maskey, M., Ganti, R., Weldemariam, K., and Ramachandran, R. (2023). Foundation models for generalist geospatial artificial intelligence. arXiv:2310.18660.

[8] Kenya Meteorological Department (2024). Climate outlook for the June-July-August 2024 season and rainfall review of the March-April-May 2024 "long rains" season. Ref. KMD/FCST/5-2024/SO/02, 31 May 2024.

[9] Konapala, G., Kumar, S. V., and Ahmad, S. K. (2021). Exploring Sentinel-1 and Sentinel-2 diversity for flood inundation mapping using deep learning. *ISPRS Journal of Photogrammetry and Remote Sensing*, 180, 163-173.

[10] Krullikowski, C., Chow, C., Wieland, M., Martinis, S., Bauer-Marschallinger, B., Roth, F., Matgen, P., Chini, M., Hostache, R., Li, Y., and Salamon, P. (2023). Estimating ensemble likelihoods for the Sentinel-1-based global flood monitoring product of the Copernicus Emergency Management Service. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16, 6917-6930.

[11] Martinis, S., Twele, A., and Voigt, S. (2009). Towards operational near real-time flood detection using a split-based automatic thresholding procedure on high resolution TerraSAR-X data. *Natural Hazards and Earth System Sciences*, 9(2), 303-314.

[12] Martinis, S., Kersten, J., and Twele, A. (2015). A fully automated TerraSAR-X based flood service. *ISPRS Journal of Photogrammetry and Remote Sensing*, 104, 203-212.

[13] Mason, D. C., Davenport, I. J., Neal, J. C., Schumann, G. J.-P., and Bates, P. D. (2012). Near real-time flood detection in urban and rural areas using high-resolution synthetic aperture radar images. *IEEE Transactions on Geoscience and Remote Sensing*, 50(8), 3041-3052.

[14] Mason, D. C., Giustarini, L., Garcia-Pintado, J., and Cloke, H. L. (2014). Detection of flooded urban areas in high resolution synthetic aperture radar images using double scattering. *International Journal of Applied Earth Observation and Geoinformation*, 28, 150-159.

[15] Mason, D. C., Dance, S. L., and Cloke, H. L. (2021). Floodwater detection in urban areas using Sentinel-1 and WorldDEM data. *Journal of Applied Remote Sensing*, 15(3), 032003.

[16] Nemni, E., Bullock, J., Belabbes, S., and Bromley, L. (2020). Fully convolutional neural network for rapid flood segmentation in synthetic aperture radar imagery. *Remote Sensing*, 12(16), 2532.

[17] Nobre, A. D., Cuartas, L. A., Hodnett, M., Rennó, C. D., Rodrigues, G., Silveira, A., Waterloo, M., and Saleska, S. (2011). Height Above the Nearest Drainage: a hydrologically relevant new terrain model. *Journal of Hydrology*, 404(1-2), 13-29.

[18] UN Office for the Coordination of Humanitarian Affairs (2024). Kenya: Heavy Rains and Flooding Update, Flash Update #7 (19 June 2024). `https://www.unocha.org/publications/report/kenya/kenya-heavy-rains-and-flooding-update-flash-update-7-19-june-2024`

[19] Otsu, N. (1979). A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1), 62-66.

[20] Ouma, C. C. (2026). Satellite-derived drought indicators as early predictors of food insecurity risk in northern Kenya's arid and semi-arid lands. Preprint. `https://craigouma.github.io/eo-drought-food-security-kenya/`

[21] Pekel, J.-F., Cottam, A., Gorelick, N., and Belward, A. S. (2016). High-resolution mapping of global surface water and its long-term changes. *Nature*, 540(7633), 418-422.

[22] Microsoft Open Source, McFarland, M., Emanuele, R., Morris, D., and Augspurger, T. (2022). *microsoft/PlanetaryComputer: October 2022*. Zenodo. `https://doi.org/10.5281/zenodo.7261897`

[23] Rennó, C. D., Nobre, A. D., Cuartas, L. A., Soares, J. V., Hodnett, M. G., Tomasella, J., and Waterloo, M. J. (2008). HAND, a new terrain descriptor using SRTM-DEM: mapping terra-firme rainforest environments in Amazonia. *Remote Sensing of Environment*, 112(9), 3469-3481.

[24] Ronneberger, O., Fischer, P., and Brox, T. (2015). U-Net: convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention, MICCAI 2015*, pp. 234-241. Springer.

[25] Salamon, P., McCormick, N., Reimer, C., Clarke, T., Bauer-Marschallinger, B., Wagner, W., Martinis, S., Chow, C., Böhnke, C., Matgen, P., Chini, M., Hostache, R., Molini, L., Fiori, E.,

and Walli, A. (2021). The new, systematic global flood monitoring product of the Copernicus Emergency Management Service. In *2021 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pp. 1053-1056.

[26] Sirko, W., Kashubin, S., Ritter, M., Annkah, A., Bouchareb, Y. S. E., Dauphin, Y., Keysers, D., Neumann, M., Cisse, M., and Quinn, J. (2021). Continental-scale building detection from high resolution satellite imagery. arXiv:2107.12283.

[27] Szwarcman, D., Roy, S., Fraccaro, P., et al. (2024). Prithvi-EO-2.0: a versatile multi-temporal foundation model for Earth observation applications. arXiv:2412.02732.

[28] Twele, A., Cao, W., Plank, S., and Martinis, S. (2016). Sentinel-1-based flood mapping: a fully automated processing chain. *International Journal of Remote Sensing*, 37(13), 2990-3004.

[29] UNITAR Satellite Centre (UNOSAT) (2024). Flood products for activation FL20240426KEN (GDACS 1102561), Kenya, 29 April to 5 May 2024. Humanitarian Data Exchange. `https://data.humdata.org/organization/unosat`

[30] Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz, S., Lesiv, M., Herold, M., Tsendbazar, N.-E., Xu, P., Ramoino, F., and Arino, O. (2022). *ESA WorldCover 10 m 2021 v200*. Zenodo. `https://doi.org/10.5281/zenodo.7254221`